

Syed Monis Sarwar

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SUMMARY

Full-stack engineer building AI-native products end to end. Three self-built products run in production, covering realtime voice pipelines, autonomous agents, and high-throughput data ingestion, each with explicit failure handling and observability. B.Tech Computer Science, graduating 2028.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL
AI & LLM: OpenAI API (Realtime, GPT-4o), Whisper, Vercel AI SDK, structured outputs, streaming tool calls, RAG over documents, OpenTelemetry tracing for LLM pipelines
Backend: Node.js, Bun, FastAPI, Inngest, Redis, BullMQ, REST APIs, webhook and queue-driven architectures
Frontend: React, Next.js, Tailwind CSS, TanStack Query, shadcn/ui
Databases: PostgreSQL, Drizzle ORM, Prisma, Neon, Supabase
Infrastructure & Tooling: Docker, Vercel, Git, pnpm, Vitest, Resend

PROJECTS

AskAI: Voice-First AI Assistant Live | [GitHub](#)

TypeScript, Next.js, OpenAI Realtime API, Stream Video SDK, Inngest, OpenTelemetry, Drizzle ORM, Neon, Vercel

- Shipped a production assistant answering questions over user documents and live meetings through the OpenAI Realtime API, with structured outputs, streaming tool calls, and explicit fallback paths for model timeouts, refusals, and malformed responses.
- Piped Stream Video SDK audio directly into the Realtime session, eliminating a full transcription round trip from the voice response path.
- Instrumented the LLM pipeline end to end with OpenTelemetry, tracing token spend, tool-call depth, and failure modes so silent model regressions surfaced as alertable metrics rather than user-reported bugs.
- Moved document ingestion and webhook fan-out off the request path into Inngest workflows with retries and idempotency keys, on serverless Postgres with Drizzle migrations.

Nivora: Autonomous Civic Grievance Agent Live | [GitHub](#)

TypeScript, Next.js, GPT-4o-mini, Whisper, Neon, Drizzle ORM, Resend

- Built an autonomous agent that classifies incoming citizen grievances, routes each to the owning department, then re-drafts and escalates the complaint up the chain with no human in the loop once an SLA window lapses.
- Kept escalation trustworthy by separating LLM judgment from a deterministic SLA state machine, so routing and timing stay auditable and replayable even when the model classifies incorrectly.
- Paired GPT-4o-mini classification with Whisper transcription to accept spoken Hindi or English, producing department-ready complaint text in the citizen's own language, with per-department compliance rates published on a public dashboard.

Curio: Content Ingestion & Ranking Platform Live | [GitHub](#)

TypeScript, Next.js, PostgreSQL, Drizzle ORM, Redis, BullMQ, Docker, Vercel

- Built an ingestion pipeline pulling **275+ sources** (RSS, Hacker News, Reddit, YouTube) with parallel per-source fetching, per-source timeouts, and fault isolation, so one failing feed never stalls or corrupts a run.
- Ranked items with a quality model across 26 topic categories to surface high-signal content over raw recency or virality.
- Shipped a timezone-aware email digest with configurable daily and weekly cadence and HMAC-signed unsubscribe links to prevent forged opt-out requests.

EDUCATION

Babu Banarasi Das Institute of Technology and Management	Lucknow, India
B.Tech, Computer Science and Engineering	2024 – 2028
Anand Bhawan School , Senior Secondary (Class XII)	2022 – 2023